

Single-Line Diagram & Equipment Schedule

Pennine BESS — 200 MW / 800 MWh to 275 kV Stalybridge

Connection point	Stalybridge 275 kV busbar
Grid transformer	200 MVA, 275/33 kV
Inverters	40 x 5 MW grid-forming
Drawing ref	PN-SLD-11, Rev B

1. Connection arrangement

The site connects to the 275 kV busbar at Stalybridge via a 200 MVA 275/33 kV grid transformer and a 33 kV switchboard. The metering point is at the 275 kV connection.

2. Power conversion

Power conversion is by 40 x 5 MW grid-forming inverters arranged in 8 blocks, each block with its own 33/0.69 kV transformer. The inverter fleet is rated for symmetric bidirectional operation at 200 MW.

3. Export limitation note

Although the grid transformer and inverter fleet are each rated to 200 MVA / 200 MW, export at the metering point is constrained by the site controller to the registered Transmission Entry Capacity. The registered TEC value should be read from the capacity-registration form, with which this diagram must be reconciled.

Equipment rating schedule

Equipment	Quantity	Unit rating	Aggregate
Grid transformer 275/33 kV	1	200 MVA	200 MVA
Inverter (grid-forming)	40	5 MW	200 MW
Block transformer 33/0.69 kV	8	25 MVA	200 MVA
Battery (LFP)	—	—	800 MWh
Site controller export cap	1	set to TEC	150 MW